

Development of MILP Model for Integrated Aircraft Maintenance Scheduling and Multi-Period Parking Layout Planning Problems

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Abstract—In this paper, two closely related problems commonly observed in aircraft hangar maintenance companies are studied: the aircraft hangar maintenance scheduling problem and the parking stand planning problem. Recently, in order to reduce operating costs in airline companies, outsourcing aircraft heavy maintenance activities to a third-party maintenance company has been gaining greater traction. In a real operation environment, aircraft with different arrival times and specific maintenance requirements from airline companies are received by an aircraft maintenance company. Hence, a maintenance schedule and a series of aircraft parking plans have to be carefully arranged to minimize the total delay over a multi-period planning horizon. However, conventional manual planning usually causes maintenance delays due to inadequate parking planning that induces blockages between arriving, departing and parked aircraft in the process of aircraft roll in and roll out operations. To deal with the inefficiency of manual planning, a Mixed-Integer Linear Programming (MILP) model that integrates maintenance scheduling and parking layout planning is proposed. To our knowledge, this is the first work to deal with the aircraft maintenance scheduling problem incorporating non-overlapping constraints between aircraft in the hangar, and is the main contribution of this research.

Keywords—aircraft maintenance scheduling; parking layout planning; mixed-integer linear programming; non-overlapping constraints

I. INTRODUCTION

Aircraft maintenance, repair and overhaul (MRO) activities are critical for the aviation industry. Maintenance checks are carried out periodically after a specified time or usage to ensure operational safety and reliability. The cost of MRO represents around 9 percent of the total annual operating cost for airline companies [1], and is the third highest cost center after fuel cost. Nowadays, operating costs with fluctuations of passenger demand has forced many airlines to reconsider their business models to ensure that their MRO operations continue to meet the safety requirements while maintaining minimum cost, maximum quality and the best lead-time [2, 3]. Outsourcing of MRO operations to an independent company has become an attractive option for airlines, especially the heavy maintenance that is conducted in a hangar. Marcontell [4] estimated that the percentage of MRO outsourcing rose 45

percent from the mid-1990s to 2012. Many airline companies have outsourced particular non-core, labour-intensive activities so as to fully focus on their high-value added commercial flying activities, such as customer relationship management and marketing activities.

From the perspective of aircraft hangar maintenance companies (also known as independent MRO service providers), fulfilling the increasing maintenance requirement of customers within a reasonable time becomes a challenging task due to limited resource availability (e.g. hangar space, staff and equipment). To fulfill aircraft maintenance orders from customers efficiently, many maintenance jobs are conducted concurrently in the hangar. Therefore, the planner has to arrange a maintenance schedule together with a series of aircraft parking plans to minimize the total delay over a multi-period planning horizon. In practice, aircraft parking stands in a maintenance hangar that fully owned by an airline company can be predefined and fixed at the hangar design stage because of the limited types of aircraft models they own, and hence aircraft are parked at the predefined stands during the maintenance process. Differing from the maintenance hangar owned by airline companies, independent hangar maintenance companies receive maintenance orders from different customers (e.g. full-cost airlines, low-cost airlines, cargo airlines, government flying service as well as private aircraft owners), which means that the maintenance company needs flexibly to arrange aircraft of different sizes with specific maintenance requirements flexibly in the hangar over the planning horizon. Therefore, defining and fixing parking stands at the hangar design stage is not suitable for maintenance companies, since the parking plan varies with the heterogeneities of incoming maintenance orders. Moreover, to reduce the risk of collision between aircraft during the maintenance process, safety considerations need to be taken into account when carrying out an aircraft parking plan.

The main contribution of this paper is a mathematical model that aids the maintenance scheduling and parking layout planning problem in the context of aircraft hangar maintenance activities. The model considers the physical shape of aircraft in the maintenance scheduling problem. Blocking between aircraft is the cause of tardiness in hangar maintenance activities. A Mixed-Integer Linear Programming model is developed to formulate the proposed problem. To our knowledge, this is the first work that

incorporates the physical shape of aircraft in the context of aircraft maintenance scheduling problem. In previous research studies, the capacity of maintenance hangar is assumed to be fixed at a constant value, which is reasonable for the hangar owned by an airline as their parking stands have been determined during the design stage. However, such an assumption is not applicable for a maintenance company since the number of aircraft they can serve varies according to the incoming maintenance requests and the physical specification of each aircraft.

II. PROBLEM DESCRIPTION AND NOTATION

This section describes the maintenance scheduling problem in the context of the maintenance hangar in the maintenance company. After introducing the assumptions and problem description, the essential constraints in the MILP formulation of the proposed problems are presented. Specifically, the methodology of preventing overlap between each pair of aircraft is firstly proposed. In addition, the utilization of hangar space is enhanced by developing a three-dimensional parking approach. Furthermore, the blocking constraints are introduced to characterize the relationships among arriving, departing and parking aircraft in each planning period to make the parking layouts accord with the maintenance schedule.

A. Problem Description and Assumptions

The maintenance scheduling problem studied in this paper originates from the hangar maintenance scheduling problem in the context of the maintenance company. In particular, the maintenance company has to carry out a maintenance schedule that records a series of aircraft parking layout plans in the hangar. Differing from the research studies in the literature that assuming the number of parking stands in hangar is fixed, we consider the parking stands are not fixed at the design stage, which corresponds to the real-life application in maintenance companies since they serve different clients and the geometric shape and specification of aircraft varies. The assumptions of this problem are as follows:

- 1) once the aircraft is rolled into the hangar, its parking position cannot be adjusted until the maintenance task is finished and the aircraft leaves the hangar;
- 2) once the aircraft is rolled into the hangar, the maintenance task must be finished before leaving the hangar;
- 3) if the arriving aircraft (or the departing aircraft) is blocked by any parked aircraft in the hangar, its movement operations cannot be conducted until its pathway is cleared; and
- 4) the moving path of an aircraft is a straight line and turning is not allowed due to the safety consideration.

B. Notations

1) Sets

a_t	Set of scheduled arrival aircraft at period t
d_t	Set of departure aircraft at period t
A_t	Set of scheduled arrival aircraft from

beginning time to period t . $A_t \in \bigcup_{i=0}^t a_i$

D_t Set of scheduled departure aircraft from beginning time to period t . $D_t \in \bigcup_{i=0}^t d_i$

A_T Set of scheduled arrival aircraft during planning period

2) Given parameters

t	period, where T is the number of periods
T^*	the end of planning period
ETA_i	Estimated time of arrival of aircraft i
ETD_i	Estimated time of departure of aircraft i
$MTime_i$	Maintenance Time of aircraft i
$penalty1_i$	Penalty of not serving aircraft i during planning period,
$penalty2_i$	Penalty of late delivery of aircraft i during planning period
$penalty3_i$	Penalty of failure to deliver aircraft i during planning period
$penalty4_i$	Penalty of late roll in of aircraft i during planning period
W	width of hangar
H	length of hangar
i	maintenance order associated with aircraft i . $i \in A_T$
w_i	width of aircraft i
h_i	length of aircraft i
TD_i	tail distance of aircraft i
NFP_{ij}	the two-dimensional No-Fit Polygon of aircraft i and j
s_{ij}^k	k th slice of the region outside the NFP_{ij}
$\alpha_{ij}^{kf}, \beta_{ij}^{kf}, q_{ij}^{kf}$	parameters used to define the f th linear equation of the slice s_{ij}^k outside the NFP_{ij}
m_{ij}	number of slices outside NFP_{ij}
t_{ij}^k	number of linear equations used to define the slice s_{ij}^k
M	large number

3) Decision variables

Continuous Decision Variables

x_i	position of reference point of aircraft i on x-axis in two-dimensional space
y_i	position of reference point of aircraft i on y-axis in two-dimensional space

Binary Decision Variables

out_{it}	1, if aircraft i is moved out at the beginning of period t
in_{it}	1, if aircraft i is moved in at the beginning of period t
out_{iT^*}	1, if fail to deliver aircraft i at the end of

	planning period
p_{it}	1, if aircraft i is parked in hangar at period t
l_{ijt}	1, if aircraft j blocks aircraft i from rolling out at period t
h_{ijt}	1, if aircraft j blocks aircraft i from rolling in at period t
lL_{ijt}	1, if aircraft i is on the left side of aircraft j without overlap at period t
lR_{ijt}	1, if aircraft i is on the right side of aircraft j without overlap at period t
lU_{ijt}	1, if aircraft i is above aircraft j without overlap at period t
hL_{ijt}	1, if aircraft i is on the left side of aircraft j without overlap at period t
hR_{ijt}	1, if aircraft i is on the right side of aircraft j without overlap at period t
hU_{ijt}	1, if aircraft i is above aircraft j without overlap at period t
b_{ijkt}	1, if the reference point of aircraft j is placed into the slice s_{ij}^k at period t

III. DEVELOPMENT OF MIXED-INTEGER PROGRAMMING MODEL (MILP)

A. Boundary and Non-Overlapping Constraints

Due to the irregular shape of aircraft, the primal consideration of the problem is to ensure that aircraft do not overlap with each other. The geometric shape of an aircraft is characterized as a non-convex polygon (Fig. 1) in two-dimensional space. Let $P = \{p_1, p_2, \dots, p_N\}$ be the set of maintenance orders that the aircraft maintenance company receives at a particular period. We denote the reference point of each aircraft as the middle point at the bottom of the aircraft, and the coordinates of the reference point of aircraft p_i in two-dimensional space are denoted as (x_i, y_i) . To prevent overlaps between two irregular polygons in two-dimensional space, the most widely used tool is the No-Fit Polygon (NFP). Bennell and Oliveira [5] provided a detailed tutorial on the approach for generating NFP between two irregular polygons. The main ideas of NFP are as follows:

For a pair of aircraft p_i and p_j , which can be characterized by two polygons, the No-fit polygon NFP_{ij} is the region in which the reference point of aircraft p_j cannot be placed if aircraft p_i remains stationary since it would overlap with aircraft p_i . The feasible zone for placing aircraft p_j without overlap with p_i is the region outside NFP_{ij} . Taking two simple polygons in Fig. 2 as an example, the NFP_{ij} is generated by tracing the path of the reference point on p_j as p_j slides around the boundary of p_i such that two polygons always touch but never overlap. Therefore, if the reference point of j moves into the NFP_{ij}

then the two polygons overlap, therefore the interior of the NFP_{ij} represents all overlapping positions. Moreover, the two polygons touch when the reference point of p_j is on the boundary of the NFP_{ij} , and they separate when the reference point is outside the NFP_{ij} .

To prevent overlap or collision, the reference point of polygon j should be placed outside the NFP_{ij} . We partition the region outside the NFP by a number of horizontal slices [6] and each horizontal slice is associated with a binary variable b_{ijkt} at period t (Fig. 3). If aircraft i and j are parked in hangar at period t , then there must be one binary variable b_{ijkt} equal to one to impose non-overlapping constraints for these two aircraft (Constraints (5) – (12)).

$$x_i + w_i / 2 \leq W, \forall i \in A_t, \forall t \geq 0 \quad (1)$$

$$x_i \geq w_i / 2, \forall i \in A_t, \forall t \geq 0 \quad (2)$$

$$y_i + h_i \leq H, \forall i \in A_t, \forall t \geq 0 \quad (3)$$

$$\alpha_{ij}^{kf}(x_j - x_i) + \beta_{ij}^{kf}(x_j - x_i) \leq q_{ij}^{kf} + M \cdot (1 - b_{ijkt}), \quad (4)$$

$$\forall i, j \in A_t, \forall k = 1, 2, \dots, m_{ij}, \forall f = 1, 2, \dots, t_{ij}^k, \forall t \geq 0$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \leq p_{it}, \forall i, j \in A_t, \forall t \geq 0 \quad (5)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \leq p_{jt}, \forall i, j \in A_t, \forall t \geq 0 \quad (6)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \leq 1 - out_{it}, \forall i, j \in A_t, \forall t \geq 0 \quad (7)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \leq 1 - out_{jt}, \forall i, j \in A_t, \forall t \geq 0 \quad (8)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \geq p_{it} + p_{jt} - 1, \forall i, j \in A_t; i, j \notin D_t; \forall t \geq 0 \quad (9)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \geq p_{it} + p_{jt} - (out_{it} + out_{jt}) - 1, \forall i, j \in A_t; i, j \in D_t; \forall t \geq 0 \quad (10)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \geq p_{it} + p_{jt} - out_{it} - 1, \forall i, j \in A_t; i \in D_t, j \notin D_t; \forall t \geq 0 \quad (11)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \geq p_{it} + p_{jt} - out_{jt} - 1, \forall i, j \in A_t; i \notin D_t, j \in D_t; \forall t \geq 0$$

$$b_{ijkt} \in \{0, 1\} \forall i, j \in A_t, k = 1, 2, \dots, m_{ij}, \forall t \geq 0 \quad (13)$$

$$p_{it} \in \{0, 1\} \forall i \in A_t, \forall t \geq 0 \quad (14)$$

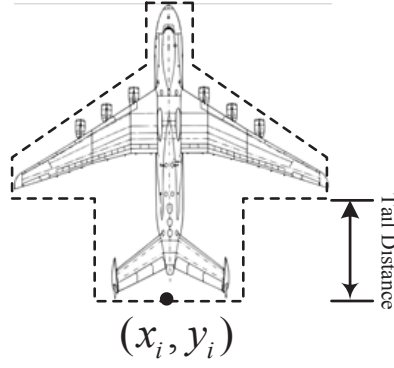


Figure 1. Geometric representation and reference point of aircraft

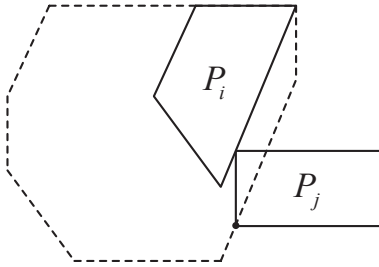


Figure 2. No-Fit Polygon of P_i and P_j

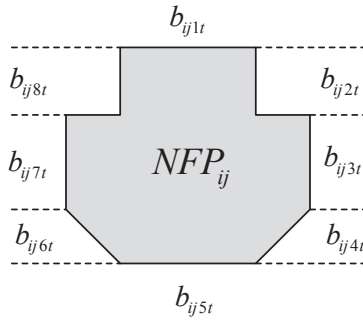


Figure 3. Horizontal Slices outside NFP

B. Three-Dimensional Aircraft Parking Plan

To further enhance the utilization of the maintenance hangar space, we find that “overlap” of aircraft wing (Fig. 4) between two aircraft is feasible as long as the two aircraft’s wings are of different heights, with a safety distance in height (Fig. 5), while the main body of aircraft should not overlap under any circumstance. In this case, another type of non-overlapping constraints can be derived by using two sets of *NFPs* that separate each pair of aircraft from three-dimensional space: The *Main Body NFP* is used to separate the main body of the two aircraft, and the *Revised Wing NFP* is used to separate the wings of aircraft with allowance of “overlap” within a safety margin. For those pairs of aircraft in different wing heights with safety margins between wings, the non-overlapping constraints (4) – (12) derived from original *NFP* can be replaced by that derived from the *Main Body NFP* and the *Revised Wing NFP*. In this connection,

two set of binary variables that act as a similar function to b_{ijkt} are introduced, and are used to separate the main body and the revised wings of aircraft, respectively.

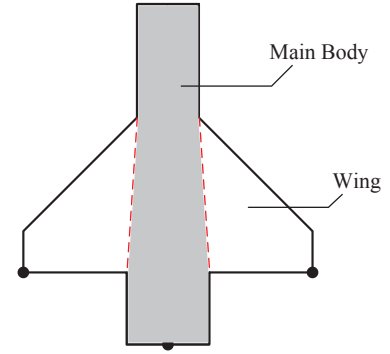


Figure 4. Parts of aircraft

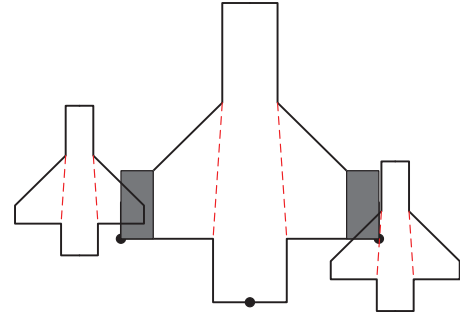


Figure 5. “Three-dimensional” parking with wings “overlap” in 2D view

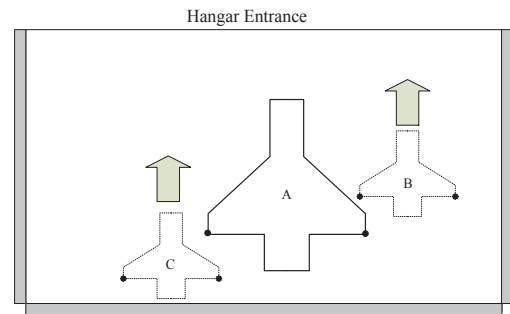


Figure 6. Roll out blocking

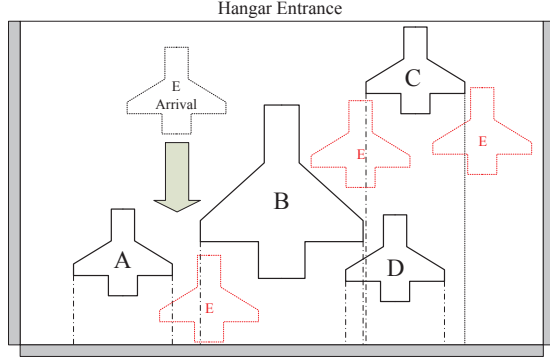


Figure 7. Roll in blocking

C. Blocking Constraints

Parking stand planning is a significant sub-problem in the proposed integrated problem since the aircraft roll in and roll out operations are mainly restricted and delayed by the insufficient hangar space and improper planning that cause series blocking in the maintenance hangar. For example, the departing aircraft cannot roll out from the hangar at the scheduled time or the arriving aircraft may not roll into the hangar in a timely manner since the other aircraft block the pathway (Fig. 6 and Fig. 7). The relationship between each pair of aircraft and the blocking constraints can be described as follows:

1) Roll out blocking

If the departing aircraft i is blocked by another aircraft j parked in the hangar, the departing aircraft i cannot roll out from the hangar in the current period t until its pathway is cleared. A set of binary variables $\{l_{ijt}, lR_{ijt}, lU_{ijt}\}$ characterizes the position relationship for pairs of aircraft placed in the hangar. If none of $\{l_{ijt}, lR_{ijt}, lU_{ijt}\}$ equals to value one, then $l_{ijt}=1$ is imposed implying that the departing aircraft i is blocked by aircraft j . As a result, an aircraft that has finished its maintenance check cannot leave the hangar at period t and the maintenance schedule has to be rescheduled in according to the parking layout.

$$(x_i + w_i / 2) - (x_j - w_j / 2) \leq M \cdot (1 - l_{ijt}) \quad \forall i \in D_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (15)$$

$$(x_i - w_i / 2) - (x_j + w_j / 2) \geq -M \cdot (1 - lR_{ijt}) \quad \forall i \in D_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (16)$$

$$(y_i + TD_i) - (y_j + TD_j) \geq -M \cdot (1 - lU_{ijt}) \quad \forall i \in D_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (17)$$

$$(1 - l_{ijt}) \geq \frac{1}{3} \cdot (lR_{ijt} + lR_{ijt} + lU_{ijt}) \quad \forall i \in D_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (18)$$

$$(1 - l_{ijt}) \leq lR_{ijt} + lR_{ijt} + lU_{ijt} \quad \forall i \in D_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (19)$$

$$l_{ijt} \leq p_{it} \quad \forall i \in D_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (20)$$

$$l_{ijt} \leq p_{jt} \quad \forall i \in D_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (21)$$

$$l_{ijt}, lR_{ijt}, lR_{ijt}, lU_{ijt} \in \{0, 1\} \quad \forall i \in D_t, j \in A_t \setminus i, \forall t \geq 0 \quad (22)$$

2) Roll in blocking

If the arriving aircraft i is about to roll into a parking position but the pathway to the target position is blocked by another aircraft j , the arriving aircraft cannot be rolled into the hangar until the pathway is cleared, or its parking position should be rearranged. Similarly, a set of binary variables $\{hL_{ijt}, hR_{ijt}, hU_{ijt}\}$ characterizes the position relationship of an arriving aircraft and a parking aircraft in the hangar. The algorithm finds a target parking position, i.e. coordinates in the hangar, for the arriving aircraft i , then a set of binary variables $\{hL_{ijt}, hR_{ijt}, hU_{ijt}\}$ determines the position relationship between them. If none of $\{hL_{ijt}, hR_{ijt}, hU_{ijt}\}$ is equal to value one, then $h_{ijt}=1$ is imposed implying that the arriving aircraft i is blocked by aircraft j to its target parking position. As a result, the arriving aircraft cannot roll into the hangar at period t with the given target parking position, and the maintenance schedule has to be rescheduled in accord with the parking layout, or rearrange the parking position for the arriving aircraft at this period.

$$(x_i + w_i / 2) - (x_j - w_j / 2) \leq M \cdot (1 - hL_{ijt}) \quad \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (23)$$

$$(x_i - w_i / 2) - (x_j + w_j / 2) \geq -M \cdot (1 - hR_{ijt}) \quad \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (24)$$

$$(y_i + TD_i) - (y_j + TD_j) \geq -M \cdot (1 - hU_{ijt}) \quad \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (25)$$

$$(1 - h_{ijt}) \geq \frac{1}{3} \cdot (hL_{ijt} + hR_{ijt} + hU_{ijt}) \quad \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (26)$$

$$(1 - h_{ijt}) \leq (hL_{ijt} + hR_{ijt} + hU_{ijt}) \quad \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (27)$$

$$h_{ijt} \leq p_{it} \quad \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (28)$$

$$h_{ijt} \leq p_{jt} \quad \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (29)$$

$$h_{ijt} \leq (1 - out_{jt}) \quad \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (30)$$

$$h_{ijt}, hL_{ijt}, hR_{ijt}, hU_{ijt} \in \{0, 1\} \quad \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (31)$$

3) Concurrent roll in & roll out

1) If more than one aircraft finish their maintenance checks and are about to leave the hangar at period t , then these two aircraft do not block each other.

$$l_{ijt} \leq 2 - (out_{it} + out_{jt}), \forall i \in D_t, \forall j \in D_t \setminus i, \forall t \geq 0 \quad (32)$$

$$out_{it} \in \{0, 1\} \quad \forall i \in A_T, \forall t \geq ETD_i$$

2) If more than one arrival aircraft are about to move into the hangar at period t , then these two aircraft do not block each other, and the aircraft going to park into the inner area of the hangar is rolled in first.

$$h_{ijt} \leq 2 - (in_{it} + in_{jt}), \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (33)$$

$$in_{it} \in \{0, 1\} \quad \forall i \in A_T, \forall t \geq ETA_i$$

4) Blocking constraints

1) If any parking aircraft blocks the move out path of the departing aircraft i , then the aircraft cannot leave the hangar at this period.

$$(1 - out_{it}) \geq \frac{1}{|A_t \setminus i|} \left(\sum_{\forall j \in A_t \setminus i} l_{ijt} \right), \forall i \in D_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (34)$$

2) If any parking aircraft blocks the path of the arriving aircraft i , then the aircraft cannot move into the hangar to the target position. As a result, the target position needs to be revised or the reschedule the move in time.

$$(1 - in_{it}) \geq \frac{1}{|A_t \setminus i|} \left(\sum_{\forall j \in A_t \setminus i} h_{ijt} \right), \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (35)$$

5) Maintenance and logical constraints

Constraint (36) imposes that the parking time of aircraft i shall be equal or larger than the required maintenance time of that maintenance request associated with aircraft i . Constraints (37) – (41) are logical constraints controlling the activation of non-overlapping constraints for aircraft i with the other aircraft parked in the hangar.

$$\left(\sum_{t \geq ETD_i} out_{it} \cdot t - \sum_{t \geq ETA_i} in_{it} \cdot t \right) + M \cdot (1 - \sum_{t \geq ETA_i} in_{it}) + M \cdot (1 - \sum_{t \geq ETD_i} out_{it}) \geq MTime_i, \forall i \in A_T \quad (36)$$

$$p_{it} = \sum_{ETA_i \leq m \leq t} in_{im}, \forall i \in A_T, \forall ETA_i \leq t \leq ETD_i \quad (37)$$

$$p_{it} = \sum_{ETA_i \leq m \leq t} in_{im} - \sum_{ETD_i \leq m \leq t-1} out_{im}, \forall i \in A_T, \forall t \geq ETD_i + 1 \quad (38)$$

$$\sum_{t \geq ETA_i} in_{it} \leq 1, \forall i \in A_T \quad (39)$$

$$out_{it} \leq \sum_{ETA_i \leq m < t} in_{im}, \forall i \in A_T, \forall t \geq ETD_i \quad (40)$$

$$(1 - out_{iT^*}) \leq \sum_{t \geq ETD_i} out_{it} + M \cdot (1 - \sum_{t \geq ETA_i} in_{it}), \forall i \in A_T \quad (41)$$

D. Objective Function

The objective (42) of the proposed maintenance problem is to minimize: the penalty cost caused by declining the maintenance request, tardiness in roll in and roll out operations, failure to delivery maintenance requests by the end of planning period.

$$\text{Minimize } \sum_{\forall i \in A_T} \left[\begin{aligned} & (1 - \sum_{t \geq ETA_i} in_{it}) \cdot \text{penalty}1_i + \\ & \sum_{t \geq ETD_i} out_{it} (t - ETD_i) \cdot \text{penalty}2_i + \\ & out_{iT^*} \cdot \text{penalty}3_i + \\ & \sum_{t \geq ETA_i} in_{it} (t - ETA_i) \cdot \text{penalty}4_i \end{aligned} \right] \quad (42)$$

IV. CONCLUSIONS

In this paper, aircraft hangar maintenance scheduling and multi-period parking layout planning problems in the context of aircraft hangar maintenance companies are studied. A Mixed-Integer Linear Programming (MILP) formulation, incorporating maintenance scheduling and parking stand planning problems, is proposed. We believe that the proposed approach assists independent aircraft maintenance companies in making the most of their limited hangar space and in increasing their service level. To the best of our knowledge, the physical shape of aircraft has not yet been incorporating in formulating the maintenance scheduling problem in the context of the hangar maintenance service providers. The proposed formulation provides an approach to enable review of the capacity of a maintenance hangar, given the variance of incoming maintenance order, from day to day. In the future, more attention will be paid on developing an efficient algorithm to solve the large-scale scheduling problem. Given the larger number of binary variables involved in the model, it is possible to decompose the original model into several sub-problems so as to reduce the complexity in solving the integrated problem.

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